

Aaron Randolph Arada

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EDUCATION

FEU Institute of Technology

Bachelor of Science in Computer Science with specialization in Software Engineering

Manila, Philippines

Aug. 2024–Present

La Salle College Antipolo

Science, Technology, Engineering, and Mathematics(STEM)

Antipolo, PH

Aug. 2022 - May 2024

PROJECTS

KUMPAS (FSL Translator) | *React, Python, TensorFlow, MediaPipe*

June 2026

- Developed a machine-learning system for real-time Filipino Sign Language recognition at ACM TechSprint: Asteria, supporting **29 gesture classes** through webcam input and MediaPipe hand landmarks
- Trained an LSTM sequence classification model with TensorFlow/Keras on **30-frame gesture sequences** containing **126 hand-landmark features per frame**
- Deployed a full-stack translator with a Vercel frontend and Railway Python inference backend, supporting translation output across **6 Philippine languages**

Aaron Intelligence | *React, FastAPI, Supabase, Gemini API*

June 2026 | [Live Demo](#)

- Engineered a RAG-based portfolio assistant indexing **18 Markdown files** into **102 searchable chunks**, enabling natural-language Q&A about projects, skills, education, and experience
- Reduced per-query context by approximately **97%** by retrieving the top **3 of 102 chunks** using Gemini Embedding 001 and Supabase pgvector instead of injecting the full knowledge base
- Integrated a React chat interface with a FastAPI backend and built a **65-question automated evaluation suite** tracking request success, retries, and median/P95 response latency

Aaron Toolkit | *React, TypeScript, FastAPI, Redis, S3, FFmpeg*

June 2026

- Architected an extensible, registry-driven utility platform where adding a tool manifest and feature module automatically updates search, navigation, and routing without homepage changes
- Built **5 utility modules** for link QR generation, YouTube/TikTok media processing, image format conversion, and text-based PDF-to-Word conversion
- Designed asynchronous processing with Redis-backed workers, S3-compatible artifact storage, pre-signed downloads, Docker Compose deployment, and Cloudflare Turnstile verification

Mirror Mentor (Hackathon Project Lead) | *FastAPI, Supabase, Gemini API, Vercel*

May 2026

- Led backend development for an AI-assisted educational platform during ACM Developers Week, designing and maintaining **17 REST API endpoints** supporting authentication, conversations, and code analysis
- Integrated Google Gemini API with a **4-phase Socratic questioning framework** to deliver guided programming feedback through multi-step code analysis workflows
- Established secure authentication with Supabase Auth and RLS, enabled real-time synchronization through Supabase Realtime, and persisted conversations in PostgreSQL; validated usability with **5 student users**

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript, Java, SQL, C++

Frontend: React, Next.js, TailwindCSS, shadcn/ui, HTML/CSS

Backend: FastAPI, Node.js, REST APIs, Supabase, PostgreSQL

AI/ML: TensorFlow/Keras, LSTM, RAG, Gemini API, Sentence Transformers, OpenCV, MediaPipe

Cloud/Tools: AWS EC2, AWS S3, Vercel, Railway, Git, DBeaver, VS Code

CERTIFICATIONS

AWS Academy Graduate - Generative AI Foundations – AWS Training and Certification, August 2026

Pearson IT Specialist: Python – Certiport, Jul. 2026

Microsoft Office Specialist: Excel Associate (Microsoft 365 Apps) – Microsoft, Aug. 2023